Veto system optimisation

Issues encountered and possible solutions

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This note collects the problems met so far in the fast veto simulation and optimisation (`Study_on_Veto_System/Optimization_Veto_System/veto`) and, for each, a possible way forward. The primary source is the open-problem tracker `ISSUES.txt` (items A1–C9, kept with their IDs, plus A6–A7 added on 4 Nov 2026); further issues were collected from `README.md`, `CHANGELOG.md`, `docs/SUMMARY_2026-10-20.md`, `docs/shield_iron_dis.md`, `docs/fast_veto_sim.md`, `docs/fastveto_physics.md`, `docs/geant4_neutrino_notes.md` and the FairShip full-simulation cross-check (items D1–D11). Rates are per spill of 5×10^{13} PoT (= kHz at 5×10^{13} PoT/s); background yields are per 15 years (6×10^{20} PoT).

Priority: A = must be fixed before optimising, B = changes the results, C = second order. **Status:** **OPEN**, **PARTLY DONE**, **DEFERRED**, **SOLVED** / **DONE** / **OK** (kept as a record of what was decided).

Overview

ID	Issue	Prio	Status
A1	Pair (combinatorial) background figure of merit saturated	A	OPEN
A2	Bounced muons only 74–77 % vetoed	A	OPEN
A3	Rate limits reported, not enforced	A	SOLVED
A4	No objective; two inconsistent cost models	A	OPEN
A5	Spill structure and PoT normalisation	A	OPEN
A6	Spectrometer field map not used: straight lines from SST1 to the timing detector	A	OPEN
A7	No predefined module dimensions; mixed-technology planes	A	OPEN
B1	Multiple scattering after the shield	B	PARTLY DONE
B2	EM debris ($e^{\pm}$, γ) from the shield end not simulated	B	OPEN
B3	Neutrino and muon DIS	B	PARTLY DONE
B4	Signal model not physical	B	OPEN
C1	Detector response simplifications	C	OPEN
C2	Dead material and dead regions	C	OPEN
C3	No validation against full simulation	C	PARTLY DONE
C4	Energy loss of bounced muons in the cavern	C	OK
C5	Weights and statistics of rare classes	C	OPEN
C6	Detector technology numbers	C	DONE
C7	V^0 from DIS outside the vessel decaying inside	C	OPEN
C8	DIS rates with the new selection	C	OPEN
C9	Unreliable bounce flag in the shah_study categories	C	OPEN
D1	Objective exploitable: signal term saturates at 10 % loss	A	OPEN
D2	Neutrino DIS in helium: irreducible for an upstream veto	B	OPEN
D3	MS7 shield DIS $\rightarrow K_L$: chain vs FairShip disagreement	B	OPEN
D4	MS7 shield DIS chain: simplifications	C	OPEN
D5	Muon source: enriched input gives 1.55$\times$ SST1 rate	B	OPEN
D6	Interaction-library approximations	C	OPEN
D7	Interactions upstream of 30.6 m (cavern concrete) not simulated	B	OPEN
D8	Results computed with the old PVC vessel wall	B	OPEN
D9	DIS$\times\mu$ pairs not redone with the V^0-stable library	C	OPEN
D10	Geant4 built-in neutrino physics unusable	—	DONE
D11	Bugs found and fixed (record)	—	DONE

A. Blockers: fix before optimising

A1 Background figure of merit is saturated — priority A — OPEN

Source: ISSUES.txt A1

Issue. The combinatorial pair background is 2–3 surviving pairs out of 196 736 candidates ($\sim 10^{-5}$ events in 15 years). Every reasonable design gives ≈ 0, so the differences between designs are Poisson noise and cannot drive an optimiser.

Possible solution.

- Score designs on the single-muon veto inefficiency per muon type, which has large statistics.
- Build the pair background by factorisation, $N_{\text{pairs}} \approx R_{\text{pairs}}\,\varepsilon_1\,\varepsilon_2$, and check that it holds on the pairs that do survive.
- Quote a 90 % CL upper limit when there are zero survivors instead of a zero.

A2 Bounced muons: the veto barely sees them — priority A — OPEN

Source: ISSUES.txt A2

Issue. Only 74–77 % of the bounced SST1 muons are vetoed. The search window contains the true crossing for only 30 % of them: a straight line extrapolated back from the spectrometer cannot follow a bounce off the cavern wall or floor. This term will probably decide the fin layout. In addition, `rock_bounced` and `veto_only` come from a single simulation run (over-fitting risk).

Possible solution.

- Study this class on its own: where it crosses the planes, before / after the bounce point (`muon_types.bounce_points`).
- Add planes downstream of the bounce point, or a search window aimed at the wall rather than along the back-extrapolated line.
- Produce more independent runs for the bounced classes before optimising on them (see C5).

A3 Rate limits are reported, not enforced — priority A — SOLVED

Source: ISSUES.txt A3; solved 4 Nov 2026 (objective.py, reco.py, bin/make_peak_maps.py)

Issue. The 4 cm tiles on the $z = 30.6$ m fins reach 10 MHz per channel (limit 1 MHz; consistent with SR-370's "up to 10 MHz in the flames"). The straw fin at 30.6 m is above 0.5 MHz/cm^2. A penalty $1 + \log_{10}(\text{peak/limit})$ existed, but grew only logarithmically above the limit, and the fin rate maps have 20×20 cm^2 bins, which average the peak down for smaller cells.

Possible solution. Done.

- Steep penalty in the objective: each limit exceeded costs $w_{\text{rate}}\,[1 + k\,(\text{peak/limit} - 1)]$, $w_{\text{rate}} = 1000$ (more than all other terms together), $k = 10$: a design $2\times$ over its limit costs 11 000 points.
- Fine peak maps: 2×2 cm^2 maps from all 506 M muons of the full dataset (UBT at 30.6 m, fins at $z = 30.6$–40.6 m, $u < 2$ m); peak over a sliding window of the channel size (pads) or 4 cm (strips), only windows with ≥ 10 effective muons; same normalisation as the coarse maps (0.1 %).
- Effect at 30.6 m: peak $\times 1.39$ (right fin 880 ± 5 vs 634 kHz/cm^2); 4 cm tiles 14.1 MHz, 2 cm tiles 3.6 MHz per channel. Penalties: D6_straw 1 103 $\rightarrow$ 12 435, D6_tile 7 578 $\rightarrow$ 253 426; E3 / E4 stay feasible.
- Left over (not A3): fibre and pixel have no rate limit yet; fins beyond 40.6 m use the coarse peak.

A4 No objective, two inconsistent cost models — priority A — OPEN

Source: ISSUES.txt A4

Issue. `optimize_sbt.py` is a stub. The cost per layer-m^2 (P-369 table 11) and the relative cost per channel (straw = 1) disagree: fibres are expensive per area, straws cheapest per channel. The pixel cost is `nan` in the area model and would break an optimiser. Free parameters and fixed limits are not written

down, and Monte Carlo noise between designs is not controlled.

Possible solution.

- Fix the objective, e.g. minimise cost (as % of the budget) subject to background $< X$ and signal loss $< Y$ (see also D1).
- Keep one cost model (area + channels combined), with a finite pixel cost.
- Write the geometric limits as constraints: cavern wall $x = -4.56$ m, 1 m straw length, nothing below $y = -3.2$ m.
- Common random numbers: fixed seed and the same events for every design.

A5 Spill and PoT normalisation — priority A — OPEN

Source: ISSUES.txt A5

Issue. Accidental hits use flat times inside a 1 s spill (`-spill`); `config.py` has 5×10^{13} PoT per spill and 6×10^{20} PoT in 15 years. The real spill has a 50 Hz ripple and a duty factor < 1; accidentals scale with the instantaneous rate, and so does the straw signal loss (5.3 % with a ± 25 ns window).

Possible solution.

- Check that all rates follow the 5×10^{13} PoT/s convention.
- Sample hit times from the measured spill profile (ripple + duty factor) and quote the signal loss at the peak instantaneous rate.

A6 Spectrometer field map not used — priority A — OPEN

Source: code (muon_types.py, reco.py, geometry.py), added 4 Nov 2026

Issue. Downstream of SST1 ($z = 82$ m) every track — muons, DIS products, V^0 and signal daughters — is extrapolated in a *straight line* to the timing detector ($z = 92.71$ m), through the spectrometer magnet ($z = 84.36$–89.56 m). The field map `MainSpectrometerField.h5` ($z = 77$–97 m) is only used for drawing. The timing-detector acceptance, the ≤ 2-tracks cut and every rate after the timing cut therefore ignore the bending: soft tracks swept out of (or into) the 2×3 m timing plane are counted wrongly, and opposite-charge pairs do not separate. The $p > 1$ GeV cut uses the true momentum.

Possible solution.

- Propagate every charged product from its state at $z = 82$ m to the timing plane through the field map (vectorised RK4, charge from the PDG code); neutrals stay straight.
- Use the propagated position for the timing acceptance and the track-multiplicity cuts.
- Optionally measure p from the bending (stations before / after the magnet, resolution model) and cut on it.
- Redo the SST1 / TD / $p > 1$ GeV rates (30.6 / 15.3 / 8.5 kHz) and compare with FairShip (48.5 / 14.6 / 7.1 kHz, field included).

A7 Detector planes have no predefined module dimensions — priority A — OPEN

Source: design.py, technology.py; module sizes from G. Frisella, 4 Nov 2026

Issue. Plane sizes are free and continuous (fin length, y range, UBT extent); channels and cost are computed for exactly that area. Only the straw (1 m) and fibre (3 m) lengths are quantised. A real plane is assembled from fixed modules, so the optimiser can propose sizes that cannot be built, and module borders are ignored. Modules to use (each carries all layers of its technology):

technology	module	content
straw	$1 \times 1\,\mathrm{m}^2$	6 layers (3 views $0, \pm 15°$), 5 mm pitch
tile2	$50 \times 50\,\mathrm{cm}^2$	$25 \times 25 = 625$ tiles of $2 \times 2\,\mathrm{cm}^2$
tile	$48 \times 48\,\mathrm{cm}^2$	$12 \times 12 = 144$ tiles of $4 \times 4\,\mathrm{cm}^2$
fibre	$0.5 \times 2.5\,\mathrm{m}^2$	two crossed SCSF-78MJ mats (x, y), 250 μm
pixel	$\sim 4 \times 4\,\mathrm{cm}^2$	quad of 2×2 Timepix4-like chips

Modules of every technology are assembled the same way: side by side (an $n_u \times n_y$ grid, e.g. 4 straw modules $= 2 \times 2\,\mathrm{m}^2$) and stacked in z at the same (x, y) (redundant stations). Technologies can be *mixed* in one plane: e.g. 1 m in x = two $50 \times 50\,\mathrm{cm}^2$ tile2 modules, or one tile2 module + one 0.5 m fibre mat, so fine timing / granularity goes only where the rate needs it. Each module has a dead frame (frame, SiPMs, readout). The module sizes are not all on one grid (tile 48 vs 50 cm, fibre 2.5 m vs straw 1 m in y): a common grid (e.g. 50 cm, tile modules padded) or mismatched borders as dead area must be decided.

Possible solution.

- `technology.py`: module size, cells per module, dead-frame width (to be fixed with the detector colleagues), channels and cost per module.
- `design.py`: a plane = a grid of module slots, each slot = (technology, number of stacked layers); these integers / choices become the optimiser's free parameters.
- `reco.py`: zero efficiency in the dead frames and gaps; frame material in the MS / DIS budget (C2).
- Rate limits (A3) per module channel; redo the existing designs on the module grid.

B. Physics missing: changes the results

B1 Multiple scattering after the shield — priority B — PARTLY DONE

Source: ISSUES.txt B1, docs/fastveto_physics.md, CHANGELOG 2026-11-04

Issue. The stored muons have no multiple scattering after the shield: the states at $z = 30.6$ m and $z = 82$ m are joined by straight lines. The material budget is now defined (air, 350 μm Mylar entrance / exit walls, 50 m helium, the veto planes): $x/X_0 = 0.0182$ without the UBT, 0.019 with a straw UBT, 0.043 with a tile UBT, i.e. $\theta_0 \approx 1.6$–2.5 mrad$/p$[GeV]. Relevant for the soft muons (1–few GeV) that dominate the fin rates. A Highland + tails model (`scattering.py`, `fastveto.msc`) is implemented and validated against Geant4 on the vessel path; still open:

- the dense planes (tile, 2.4 % X_0) are not checked against Geant4;
- the material of the vessel *side* walls (muons entering through the SBT) is not defined;
- for bounced muons MS must start only after the bounce point.

Possible solution.

- Run `bin/msc_validate.py` with a tile plane in the path.
- Define the side-wall material and add it to the per-segment walk (A→K→B).
- Check that `RecoConfig.ms_slope` / `r_ms` (window growth with $1/p$) still cover the true scatter; otherwise the window loses efficiency or gains accidentals.

B2 EM debris from the end of the shield — priority B — OPEN

Source: ISSUES.txt B2

Issue. Only muons are simulated: no δ rays, bremsstrahlung or pair-production debris leaking from the last iron of the shield. These add hits (occupancy, rate limits) and accidental vetoes (signal loss). The FairShip full-physics cross-check shows how large this can be: all charged particles at SST1 are ~ 3.6 MHz vs 48.5 kHz of muons, dominated by $e^{\pm}$.

Possible solution.

- From the full dataset (506 M rows at $z = 30.6$ m), compute per muon the probability to produce an

escaping $e^{\pm}$ / γ in the last 20 cm of iron ($z = 29.69$–29.89 m, 11.4 X_0, equilibrium layer): δ rays ($dN/dT \simeq 0.56$ MeV cm$^{-1}/T^2$), bremsstrahlung, direct pair production.

- Turn it into e/γ rate maps on the UBT and fins, apply a per-technology detection probability (photons rarely convert in thin straws, more in 1 cm tiles), add to the accidentals.
- Cross-check with Geant4 (muon beam on 20 cm iron) and with the FairShip EM sample.

B3 Neutrino and muon DIS — priority B — PARTLY DONE

Source: ISSUES.txt B3, README, docs/fast_veto_sim.md

Issue. Originally deferred. Since then forced muon DIS (Geant4 + Pythia 6 library) and neutrino interactions (GENIE 3.06.02) are modelled in helium, air, vessel walls, every plane and the last 50 cm of MS7. Still missing: DIS in the cavern (see D7) and the material term in the optimiser — without it the optimiser favours thick, fast planes for free (tiles 2.4 % X_0 vs straws 0.12 %).

Possible solution.

- Keep the per-plane interaction source in the objective (already in `objective.score`) so plane material is penalised through its own DIS.
- Update the status in `ISSUES.txt` (still says DEFERRED).

B4 Signal model — priority B — OPEN

Source: ISSUES.txt B4

Issue. The signal is a neutral particle decaying uniformly in the vessel, two daughters uniform over the SST1 acceptance, p uniform 1–50 GeV. Signal is lost only through accidental hits. Opening angles, momenta and the vertex distribution are not realistic.

Possible solution.

- Use HNL and dark-photon models from the literature: decay probability $P = e^{-L_1/\lambda} - e^{-L_2/\lambda}$, $\lambda = \beta\gamma c\tau$; production from charm/beauty at 400 GeV; channels $\mu\pi$, $e\pi$, $\mu\mu\nu$; 2/3-body kinematics. Take them from the FairShip generator or [1–5], not invented.
- A truth-level path exists: the collaboration LLP generator EventCalc (per-channel cut efficiencies already studied in `EventCalc_signal_rejection`); feed its decays through the veto.

[1] SHiP, JHEP 04 (2019) 077; [2] Bondarenko et al., JHEP 11 (2018) 032; [3] SHiP, EPJC 81 (2021) 451; [4] Alekhin et al., Rep. Prog. Phys. 79 (2016) 124201; [5] CERN-SPSC-2023-033.

C. Detector and bookkeeping: second order

C1 Detector response simplifications — priority C — OPEN

Source: ISSUES.txt C1

Issue. σ_x is not used (the 5 cm search window dominates); no ghost hits from combining views when several muons share a time window; straw time = uniform smearing over the 50 ns drift window; no cluster size, no tile cross-talk; plane efficiencies independent.

Possible solution. Add ghost-hit combinatorics per time window (they matter at the fin rates); with a tile t_0 use the drift radius for the straw time and position; correlated inefficiencies (module-level) as a systematic.

C2 Dead material and dead regions — priority C — OPEN

Source: ISSUES.txt C2

Issue. No frames, manifolds, overlaps between 1 m straw rows, module gaps, SiPMs or cables. For the UBT this matters most: the tensioned-straw frame sits in the acceptance (SR-370 open question).

Possible solution. Add the UBT frame and module gaps as passive material (MS + DIS target) and as dead regions (zero efficiency); measure the inefficiency of the gaps with the single-muon figure of merit (A1).

C3 Validation against full simulation — priority C — PARTLY DONE

Source: ISSUES.txt C3, FairShip EM SST1 cross-check

Issue. The fast simulation was never run end-to-end against FairShip for one design. A first full-physics check (FairShip `TRY_2026_EM_SST1_origin`, 411.6 M muons) gives muons at SST1 / TD / $p > 1$ GeV of 48.5 / 14.6 / 7.1 kHz vs 30.6 / 15.3 / 8.5 kHz in the fast chain; the excess after the first cut is π decays in the vessel. Shield-made hadrons disagree strongly (see D3).

Possible solution. Run one baseline design through FairShip and compare rates per plane and the veto inefficiency per muon type before trusting an optimum.

C4 Energy loss of bounced muons in the cavern — priority C — OK

Source: ISSUES.txt C4

Issue. Whether the energy loss in the cavern rock is modelled for the bounced muons.

Possible solution. Nothing to do: it is in the stored muons from the transport.

C5 Weights and statistics — priority C — OPEN

Source: ISSUES.txt C5, CHANGELOG 2026-10-30

Issue. The repeated-`evt_n` weight correction is not applied in the optimisation dataset; `veto_only` and `rock_bounced` come from one run. For muon DIS in helium one input muon gives 6.8×10^4 of 10^6 draws: $n_{\text{eff}} \approx 60$, ± 23 % on the rate. Flattening the weights to 1 was rejected: the $w = 768.75$ minimum-bias class is 40.5 % of the true shield DIS rate.

Possible solution. Decide on the `evt_n` correction; one forced interaction per crossing / per track is already used (errors halved for ν); produce more independent seeds for the rare classes, and keep bootstrapping over `evt_n` for all errors.

C6 Detector technology numbers — priority C — DONE

Source: ISSUES.txt C6 (commit f6d13ec)

Issue. Straw / fibre / pixel / tile parameters needed the colleagues' input.

Possible solution. Updated in `technology.py`. Still to confirm: fibre timing (0.5 ns kept, may reach ~ 100 ps), per-SiPM efficiency, tile channel count for mixed tile sizes.

C7 V^0 from DIS outside the vessel decaying inside — priority C — OPEN

Source: ISSUES.txt C7

Issue. A K_S / Λ made by muon DIS in cavern air, a fin or the UBT can decay inside the vessel with no upstream hit of its own. Air: 0 of 45 air pairs after DOCA have a vertex in the vessel, but one air draw corresponds to 7×10^6 interactions per 15 y (not conclusive). Fins / UBT are not in the interaction caches; K_L kept stable.

Possible solution. Draws restricted to air and fins close to the vessel entrance (`dis_cutflow.py`); add the fins / UBT to the interaction caches; decay K_L in flight (as already done for MS7, D3).

C8 DIS rates with the new selection — priority C — OPEN

Source: ISSUES.txt C8

Issue. Without the measured IP < 10 cm and ≤ 2 tracks in SST1, before the veto per 15 y: μ He $(6.0 \pm 1.4) \times 10^4$, μ wall $7\,400 \pm 850$ (old PVC), ν He 467 ± 2 (old selection: 1 800 and 0.82).

Possible solution. Decide whether to bring back a pointing / track-multiplicity cut; the daughter-asymmetry cuts ($|A_p|, |A_{IP}| < 0.9$) keep 48 % of μ He DIS and 99 % of the wall DIS, so they alone are not enough. Redo the wall numbers with Mylar (D8).

C9 shah_study categories: bounce flag not reliable — priority C — OPEN

Source: ISSUES.txt C9, CHANGELOG 2026-11-04

Issue. In `cuda_muons_ship.categorize_muons` every `bounced_back` muon checked (seed 0) is a perfect straight line from 30.6 m to 82 m, and `straight_past_vessel` excludes straight muons through the vessel and contains 0.6 % bounced ones. With the category the straight & selected UBT rate was 6.24 kHz instead of 8.13 kHz.

Possible solution. Use the veto definition (offset from the 30.6 m line $<$ `BOUNCE_CUT` = 0.5 m, `muon_types.py`); report the bug to the MuonsAndMatter authors.

D. Further issues from the other documents

D1 Objective exploitable: signal term saturates — priority A — OPEN

Source: docs/SUMMARY_2026-10-20.md §5–6

Issue. reward $= -[100(1-e^{-N}) + 10\min(\text{sig. lost}/0.10, 1) + \text{cost}]$. The signal term saturates at 10 % loss, so losing 10 % and 74 % cost the same. Once neutrinos dominate N, vetoing everything pays: evolution run 3 reached its best reward with a design losing 74 % of the signal (50 mrad cone, ~ 15 planes) to bring ν from 0.6 to 0.06.

Possible solution.

- A sensitivity figure of merit, e.g. maximise $\varepsilon_{\text{sig}}/N_{90\%\,\text{UL}}$.
- Leave neutrinos out of N in the optimiser (or count them before the veto), since an upstream veto cannot remove them (D2).
- A linear, uncapped signal term with a hard penalty above ~ 5 % loss.

D2 Neutrino DIS in helium: irreducible for an upstream veto — priority B — OPEN

Source: docs/SUMMARY_2026-10-20.md §4

Issue. About 0.6 ν DIS in the helium per 15 y survive all designs: the vertex is in the gas, nothing charged comes in, and the third that is vetoed is vetoed by random hits (like signal). Tagging the event's own extra products (wide-angle hadrons hitting fins downstream of the vertex, SBT-style) is not modelled: `interactions.charged_tracks` keeps only forward tracks and the veto searches only upstream.

Possible solution. Keep all charged / neutral final-state particles, digitise hits downstream of the vertex on the fins, and add a "vertex-activity" veto; evaluate its signal cost with the realistic signal model (B4).

D3 MS7 shield DIS $\to K_L$: chain disagrees with FairShip — priority B — OPEN

Source: docs/shield_iron_dis.md, FairShip EM SST1 check

Issue. The forced chain in the last 50 cm of MS7 gives K_L decays in the vessel of ~ 4.9 kHz and $(6.73 \pm 0.59) \times 10^7$ per 15 y after DOCA, fiducial and IP < 2.5 m. Concerns:

- unforced Geant4 gives only 0.52 ± 0.15 of the chain's K_L; the excess comes from the Pythia 6 component of `mu_library_iron` (9.5 % of interactions, 90 % of the K_L);

- FairShip full physics sees *zero* K_L/K_S from the shield reaching the SST, and shield-made π are $\sim 17\times$ below the chain — not explained by the Pythia excess;
- after the full selection the μ number rests on 3–7 MC events (± 40–60 %);
- algorithm A cannot veto it (decay tracks have no upstream hits); asymmetry cuts keep 100 %.

Possible solution. Find the source of the FairShip discrepancy: MS7 flux and field TRY6 vs TRY2026, the muon-nuclear model (Kokoulin in FairShip vs Pythia 6 high-ν DIS here), the flux footprint already agrees. Use K_L-enriched draws for statistics. Model a veto on any in-time hit (scattered muon in the UBT / fins) and a pointing cut.

D4 MS7 shield DIS chain: simplifications — priority C — OPEN

Source: docs/shield_iron_dis.md (Limitations)

Issue. Straight parent lines from 29.20 m (no bending in 2 T, ~ 1.4 cm at 20 GeV); muons stopping in the slab not removed; Pythia part on free nucleons; straight lines and no field after the plane; π/K decays in flight ignored; neutron and K_L interactions in the vessel / air not simulated; neutrino flux $p > 10$ GeV only, no ν_e.

Possible solution. Propagate the parent with the field map (already in `shield_iron.py`); decay π/K in flight and transport neutrons / K_L in helium and air; extend the neutrino flux below 10 GeV and add ν_e.

D5 Muon source: enriched input disagrees with the old sample — priority B — OPEN

Source: docs/fast_veto_sim.md §5

Issue. The SST1-selected rate is $1.55\times$ higher with the enriched input ($1.9\times$ above 100 GeV) at the same flux above 150 GeV — larger than any statistical error. In addition the regeneration differs by 10–20 % from the old file, the > 200 GeV tail has a $\sim \times 2$ model dependence (Pythia version / diffraction), and 50–150 GeV still has the old statistics.

Possible solution. Compare angle and momentum shapes of the two inputs above 150 GeV; extend the enrichment down to 50 GeV; quote the source model as a systematic on the muon backgrounds.

D6 Interaction-library approximations — priority C — OPEN

Source: docs/fastveto_physics.md (Open)

Issue. Final states are not rescaled to the exact muon energy (15 % grid); no nuclear effects (Fermi motion, shadowing, re-interactions) in the Pythia part; no τ library (~ 1 % ν_τ in the flux).

Possible solution. Rescale final states to the true energy (or finer grid); compare with a Geant4/GENIE nuclear model on a subset; build a ν_τ library.

D7 Interactions upstream of 30.6 m not simulated — priority B — OPEN

Source: docs/fast_veto_sim.md §5, FairShip origin breakdown

Issue. Apart from the last 50 cm of MS7, interactions in the shield, cavern walls and rock are not simulated. FairShip shows that π / p at SST1 come mostly from the cavern concrete and the vessel walls / SBT.

Possible solution. Add the cavern concrete (and the rest of the shield) as forced-DIS targets using the bounced-muon paths; use the FairShip origin breakdown as the reference.

D8 Results computed with the old PVC vessel wall — priority B — OPEN

Source: CHANGELOG 2026-11-04

Issue. The vessel wall is now 350 μm Mylar (x/X_0 1.53 $\rightarrow$ 1.23 $\times 10^{-3}$, +18 % nucleons, no Cl). `results/interactions`, apply / reco-input files with “pvc” rows, scattering-dependent reco and the C8

wall numbers are still PVC-based.

Possible solution. Rerun the wall DIS and the reco with Mylar; rename the target key `"pvc"` to `"wall"` once all saved files are regenerated.

D9 DIS×μ pairs not redone — priority C — OPEN

Source: shield DIS notes (2026-11-02)

Issue. With K_S/Λ now tracked in the MS7 iron (`mu_library_iron_v0stable.npz`), the `dis_comb_*` DIS×μ combinatorial pairs were not redone.

Possible solution. Rerun `dis_comb_pairs_asym.py` on the V^0-stable sample.

D10 Geant4 built-in neutrino physics unusable — priority — — DONE

Source: docs/geant4_neutrino_notes.md

Issue. In Geant4 11.2.2 all ν–nucleus models share one set of x/Q^2 tables (CC sampled from NC kinematics, $\langle y \rangle = 0.13$–0.34 instead of 0.48), NC is $2.6\times$ too large and drops all hadrons, no Lund fragmentation, and a `fWeight` bug gives CC secondaries weight 1.

Possible solution. Done: GENIE 3.06.02 (tune G18_02a) built and used for all neutrino interactions.

D11 Bugs found and fixed (record) — priority — — DONE

Source: CHANGELOG, SUMMARY 2026-10-20

Bugs that changed results, with the fix:

- `reco.own_hits`: with MS on, tracks born downstream got a UBT hit at their start (9.5 % fake vetoes for ν DIS in helium) — start-upstream condition restored (2026-11-01).
- `make_design_explorer.py`: muons stopping in the shield (type 7) counted on the UBT (5.3×10^9 instead of 1.05×10^7 per spill) — excluded.
- `shield_dis`: muon final states sampled twice per interaction — fixed.
- shah_study “straight” category replaced by the `BOUNCE_CUT` definition (C9).